



Reichman University

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Final Project Report

Permuted Dogs versus Cats

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Abstract

This project studies binary Dogs versus Cats classification after fixed tile-wise permutations are applied to each image. The proposal frames the task as a relaxed image-domain analogue of permuted-puzzle problems with cryptographic motivation: instead of pixel-wise permutations over function graphs, images are divided into coarse grids and their tiles are rearranged before classification. We implement a notebook-first experimental pipeline around the Kaggle Dogs vs Cats dataset, train baselines on unpermuted and permuted images, evaluate augmentation, curriculum, mixed-input, and classifier-head strategies, and define model-agnostic permutation hardness metrics. The final experiment matrix uses four grid settings, 1 tile / none, 4-by-4, 7-by-7, and 10-by-10, crossed with three deterministic permutation functions named easy, medium, and hard. Part 1 compares frozen-backbone ResNet-18, DeiT-Tiny, and MLP-Mixer models; Part 2 evaluates ResNet-18 improvement ablations; and Part 3 joins the deterministic permutation metrics to ResNet-18 validation accuracy over all 12 conditions.

1 Introduction

Image classification models are typically evaluated on data whose spatial structure is intact. This project asks what happens when that structure is deliberately disrupted before learning. Each input image is split into an $N \times N$ grid, the tiles are permuted by a fixed mapping, and the classifier must still distinguish dogs from cats. The task is motivated by a proposal for an intermediate experiment connected to permuted puzzles and cryptographic hardness [1]. The image setting relaxes the original cryptographic problem in two ways: it uses natural images rather than graphs of functions, and it applies tile-wise permutations rather than pixel-wise permutations.

The project has three main objectives. First, it measures how tile resolution affects the feasibility of classification after permutation. Second, it asks whether a neural network succeeds by learning or compensating for the permutation, or whether enough class-discriminative signal remains directly visible in the permuted images. Third, it develops ways to rank permutations according to their induced difficulty for a fixed dataset and tile count.

The implementation follows a reproducible, notebook-first workflow. Part 1 trains model baselines on unpermuted images and deterministic tile permutations at 4-by-4, 7-by-7, and 10-by-10 resolution. Part 2 tests improvement strategies for the same permuted setting, including patch shuffle, regular image augmentation, mixed original/permuted training, a larger ResNet-18 classification head, and two curricula. Part 3 computes permutation hardness metrics and joins them to validation accuracy so that image-independent permutation properties can be compared against model performance.

2 Related Works

The motivating proposal cites permuted puzzles as a source of cryptographic hardness, where fixed permutations can make structured objects difficult to recover or classify [1]. This project does not attempt to solve the cryptographic problem directly. Instead, it uses the Dogs vs Cats setting as an empirical stepping stone for studying how much semantic information survives structured spatial disruption.

Convolutional neural networks are a natural starting point because they exploit locality and translational structure in images. ResNet architectures are especially relevant because residual connections improve optimization of deep convolutional models [3]. Since tile permutations partially destroy local adjacency while preserving tile content, ResNet-18 provides a useful baseline for testing whether standard convolutional inductive biases remain helpful.

The ablations also connect to standard regularization and robustness ideas in deep learning [2]. Patch shuffling and regular image augmentations expose the model to additional nuisance variation, mixed original/permuted training tests whether clean-layout examples help the permuted task, and a small MLP classifier head tests whether extra trainable head capacity helps when the backbone is frozen. Curriculum learning variants gradually control corruption probability or permutation difficulty, testing whether staged exposure to harder permuted inputs improves validation accuracy.

3 Data Overview

The dataset is Kaggle’s Dogs vs Cats image dataset, with labels inferred from filenames such as `cat.123.jpg` and `dog.456.jpg`. The

repository expects labeled training images under `data/dogs-vs-cats/train`; the unlabeled Kaggle test folder is not used for validation accuracy. The task is binary classification, and the saved experiment outputs report validation accuracy after the configured training runs.

For the unpermuted setting, each image is processed as a normal dog or cat image and used as the benchmark for post-permutation performance. For the permuted setting, each image is divided into a square tile grid and the same deterministic tile permutation is applied across images for a run. The final grid settings are one tile / none, 16 tiles, 49 tiles, and 100 tiles. The one-tile baseline is repeated under the easy, medium, and hard labels so that all parts use the same 4×3 grid-permutation matrix.

4 Methodology

The experimental pipeline is organized into three notebooks. Part 1 establishes baselines by training on the unpermuted task and on fixed deterministic tile permutations. Part 2 reuses the Part 1 ResNet-18 baseline and trains improvement ablations on the same task family. Part 3 computes permutation descriptors and correlates them with the validation accuracy observed for the corresponding deterministic records.

The primary Part 2 and Part 3 model is ResNet-18 [3]; Part 1 also compares DeiT-Tiny and MLP-Mixer. Experiments save raw run outputs, aggregated CSV summaries, and figures under `outputs/results` and `outputs/figures`. Aggregation reports mean final validation accuracy, mean best validation accuracy, standard deviation across the three deterministic permutation labels at each tile count, and the number of runs.

4.1 Training and Validation Sets by Experiment

All experiments use the labeled Kaggle Dogs vs Cats training folder as the only source of supervised examples. Labels are inferred from filenames, and the samples are split once with a stratified class-balanced split using seed 42, validation fraction 0.2, and test fraction 0.0. For the full dataset this gives 20,000 training images and 5,000 validation images: 10,000 cats and 10,000 dogs for training, and 2,500 cats and 2,500 dogs for validation. The validation split is held out from optimization and is not modified by training-only augmentations such as patch shuffle, regular augmentation, original/permuted sampling, or curriculum-stage sampling.

The curriculum ablations in Part 2 use stage-specific training loaders, but they do not create separate held-out validation splits per stage. Across both curricula, validation remains the same 5,000-image held-out loader for the final target run.

4.2 Deterministic Permutation Functions

The final experiments no longer sample random tile permutations for the main matrix. Instead, every part uses one shared deterministic generator with three parameter presets: easy, medium, and hard. This makes comparisons across Parts 1, 2, and 3 reproducible and interpretable, because a given difficulty label refers to the same construction wherever it appears.

The generator is controlled by relative parameters: the fraction of tiles to swap, the maximum swap distance as a fraction of grid width, row and column shift fractions, and whether to apply a global reversal before the shifts and swaps. This matters because fixed operation counts are not comparable across

Part	1	Training set	Validation set
run			
1 tile / none		20,000 unpermuted images repeated under easy, medium, and hard labels.	5,000 unpermuted held-out images.
4-by-4 permutations		Same 20,000 images with easy, medium, and hard deterministic 4-by-4 permutations.	Same 5,000 held-out images with the matching deterministic permutation.
7-by-7 permutations		Same 20,000 images with easy, medium, and hard deterministic 7-by-7 permutations.	Same 5,000 held-out images with the matching deterministic permutation.
10-by-10 permutations		Same 20,000 images with easy, medium, and hard deterministic 10-by-10 permutations.	Same 5,000 held-out images with the matching deterministic permutation.

Table 1: Part 1 training and validation sets. The same split is reused for ResNet-18, DeiT-Tiny, and MLP-Mixer Base, and each remote run trains for 10 epochs.

grid sizes: two swaps affect 25% of the tiles in a 4-by-4 grid but only 4% of the tiles in a 10-by-10 grid. The easy preset therefore uses a small tile-count-scaled swap fraction and short local movement; medium increases the swap fraction and adds moderate row and column shifts; hard uses the largest swap fraction, broader movement radius, stronger shifts, and a global reversal. For the one-tile baseline, all three labels map to the unpermuted image; the labels are still saved so the experiment matrix remains exactly 4 grid settings by 3 presets.

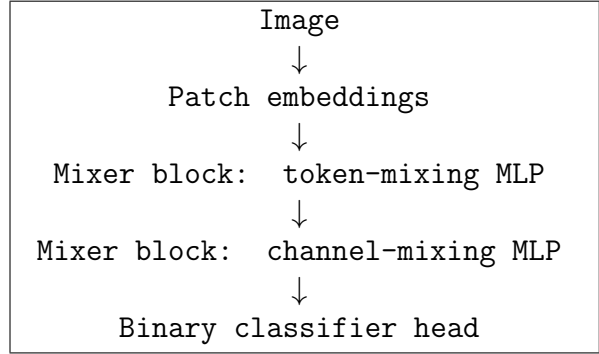


Figure 1: Simplified MLP-Mixer flow. Token mixing exchanges information across patches, while channel mixing recombines features within each patch representation.

4.3 Architecture Comparison and Frozen Backbones

To compare architectural inductive biases beyond ResNet-18, the Part 1 pipeline also supports three pretrained feature extractors: ResNet-18, DeiT-Tiny [5], and MLP-Mixer [4]. All three are used in the same transfer-learning regime: the ImageNet-pretrained backbone is frozen and only the final binary classification head is optimized. This makes the relevant training-capacity comparison the number of learnable head parameters, not the total number of parameters stored in the network.

The three models differ mainly in how they exchange information across the image. ResNet-18 uses convolutions, giving it a strong local-neighborhood bias. DeiT-Tiny represents the image as patches and uses self-attention to let patches interact. MLP-Mixer also operates on image patches, but replaces attention and convolutions with two MLP operations: token mixing, which combines information across patches, and channel mixing, which recombines feature dimensions inside each patch representation.

The MLP-Mixer choice required special care. The smaller `mixer_s16_224` archi-

texture would have been closer in total parameter count, but the available `timm` configuration does not provide pretrained weights for that variant. Freezing an untrained MLP-Mixer backbone would train only a classifier on random fixed features, which would not be a meaningful comparison to pretrained ResNet-18 and DeiT-Tiny. We therefore use the larger pretrained `mixer_b16_224.goog_in21k_ft_in1k` model. Although this increases the number of frozen parameters and may increase forward-pass cost, the number of optimized parameters remains comparable to the other models: 1,538 learnable parameters for MLP-Mixer Base versus 1,026 for ResNet-18 and 386 for DeiT-Tiny.

4.4 Improvement Strategies

Part 2 evaluates training strategies against the regular Part 1 ResNet-18 baseline. The main distinction is whether a strategy adds stochastic training views, mixes clean and permuted inputs, changes the classifier head, or changes the order in which the model sees easy and hard inputs. In all cases, validation is performed on the held-out validation images with the target permutation and without training-only perturbations.

For the counts below, let B be the configured batch size and let $N_{\text{train}} = 20,000$ be the number of training images. The ablations do not append files to the dataset. Instead, they replace or resample the ordinary training view during the forward pass, so the stored increase is zero even when every processed example is augmented or curriculum-scheduled.

4.4.1 Regular Part 1 Baseline

Main idea. This is the reference point for Part 2. ResNet-18 is trained normally on the target permutation, with no extra augmentation, no curriculum, and the standard loss.

Reason. The baseline is necessary because every ablation must be judged against ordinary training on the same task. Without it, an improvement strategy might look useful simply because the target permutation is easier than another run.

Deeper dive. The baseline uses the same train and validation split as Part 1. For a permuted run, both training and validation images are transformed by the same fixed tile permutation for that run. It has one training stage and does not increase the dataset size. If x_i is an input image, P_π is the target tile permutation, and f_θ is the classifier, the baseline minimizes

$$\frac{1}{N_{\text{train}}} \sum_{i=1}^{N_{\text{train}}} \ell(f_\theta(P_\pi(x_i)), y_i).$$

The number of added augmented images is 0 per batch and 0 per epoch; the number of processed training views remains B per batch and 20,000 per epoch.

4.4.2 Patch Shuffle

Main idea. Patch shuffle randomly rearranges smaller patches during training. The model therefore sees extra local spatial disruptions in addition to the fixed tile permutation.

Reason. This is relevant because the main task already tests whether class information survives spatial reordering. If extra random patch movement helps, it suggests that robustness to local layout disruption is useful for permuted Dogs versus Cats classification.

Deeper dive. The target permutation is still the run-level task definition, but each training batch can also receive random patch shuffling. Let S_ρ denote a random within-image patch permutation sampled independently during training. The effective training

view is

$$\tilde{x}_i = S_{\rho_i}(P_{\pi}(x_i)),$$

and the loss is computed on $\ell(f_{\theta}(\tilde{x}_i), y_i)$. In the implementation the shuffle probability is 1.0, so a batch of size B contains B shuffled views and an epoch contains 20,000 shuffled views. These views are not appended to the dataset; they replace the current batch examples. The validation loader does not use patch shuffle, so reported accuracy measures the fixed target validation permutation.

4.4.3 Regular Augmentations

Main idea. Regular augmentations apply standard image-level stochastic transformations during training while keeping the target validation permutation fixed.

Reason. This tests whether generic image augmentation helps the frozen ResNet-18 head learn more robust dog-versus-cat cues under tile permutation, or whether the extra variation is mismatched to the spatial corruption used at validation time.

Deeper dive. The training set remains the same 20,000 images. For each example, an image-level transform E is sampled during training:

$$\tilde{x}_i = E(P_{\pi}(x_i)).$$

The strategy has one stage and trains for 10 epochs. It stores no additional images; it produces one stochastic view per processed training example. Validation uses the deterministic target permutation without training-only augmentation.

4.4.4 Mixed Original/Permuted Training

Main idea. Mixed original/permuted training exposes the model to clean images and target-permuted images in the same run.

Reason. If clean images preserve class structure that is useful for the permuted task, mixing them into training may stabilize the classifier. If the clean examples pull the head away from the target validation distribution, the strategy may hurt.

Deeper dive. The ablation samples each training view as original with probability $p_{\text{original}} = 0.5$ and target-permuted otherwise:

$$\tilde{x}_i = \begin{cases} x_i, & z_i < p_{\text{original}}, \\ P_{\pi}(x_i), & z_i \geq p_{\text{original}}, \end{cases}$$

where $z_i \sim \text{Uniform}(0, 1)$. This gives an expected 10,000 clean views and 10,000 permuted views per epoch. The validation loader remains fully target-permuted, so the reported accuracy measures transfer from the mixed training distribution to the target permutation.

4.4.5 ResNet-18 MLP Head

Main idea. This ablation keeps the frozen ResNet-18 backbone but replaces the linear classifier with a small MLP classification head.

Reason. The frozen backbone may still extract useful ImageNet features from permuted tiles, but the mapping from those features to dog/cat labels may be less linearly separable after spatial disruption. A small MLP head tests whether additional trainable capacity at the classifier stage improves the target task.

Deeper dive. The data, target permutations, validation loader, and 10-epoch non-curriculum schedule match the regular Part 1 baseline. The only intentional change is the classifier head, so improvements can be interpreted as evidence that head capacity matters under the frozen-backbone setup.

4.4.6 Corruption-Probability Curriculum

Main idea. The model starts with mostly intact images and gradually sees the target permutation more often. The corruption probability increases across four stages.

Reason. Jumping immediately into a hard permutation may make optimization unstable. This curriculum tests whether gradual exposure lets the model first learn dog-versus-cat cues and then adapt those cues to more frequent spatial corruption.

Deeper dive. Each stage uses the same 20,000 training images and target tile permutation, but changes the probability that the permutation is applied. For stage probability q_s , the sampled training view is

$$\tilde{x}_i = \begin{cases} P_\pi(x_i), & z_i < q_s, \\ x_i, & z_i \geq q_s, \end{cases}$$

$$z_i \sim \text{Uniform}(0, 1).$$

The stages use $q_s \in \{0.1, 0.3, 0.5, 0.8\}$, so the expected number of permuted views per stage epoch is $q_s N_{\text{train}}$: 2,000, 6,000, 10,000, and 16,000 respectively. In a batch of size B , the expected number of permuted views is $q_s B$. This does not store a larger dataset; it changes the sampled training view. There is no separate validation set for each stage; validation always uses the fixed target validation loader. Curriculum runs train for 30 epochs in the final saved results.

4.4.7 Permutation-Difficulty Curriculum

Main idea. The model starts on easier spatial structure and moves toward the final harder permutation. Instead of changing probability, it changes the tile grid difficulty across stages.

Reason. Smaller or easier permutations preserve more of the original image structure.

This experiment tests whether a model can learn useful class cues under mild disruption before facing the final, harder tile rearrangement.

Deeper dive. Each stage uses the same 20,000 training images but changes the tile grid used for training. If g_s is the stage grid size and π_s is the corresponding sampled or target permutation, then

$$\tilde{x}_i^{(s)} = \begin{cases} x_i, & g_s = 1, \\ P_{\pi_s}^{(g_s)}(x_i), & g_s > 1. \end{cases}$$

The schedule begins with original images and then progresses through 2-by-2, 3 tiles per side, and 4-by-4 permutations; for 7-by-7 and 10-by-10 target runs, it adds a final target-grid stage. Stage permutations use the same deterministic difficulty label as the target record. Each stage processes B scheduled-difficulty views per batch and 20,000 per stage epoch. The stored dataset size does not increase, validation remains the held-out loader for the final target run, and curriculum runs train for 30 epochs in the final saved results.

4.5 Tile Permutation Hardness Metrics

We quantify the difficulty induced by a tile permutation using three complementary, model-agnostic axes and a fourth combined score. Let the image be divided into an $N \times N$ grid. For each source tile i , let (r_i, c_i) denote its original row and column, and let (r'_i, c'_i) denote its destination after applying the permutation.

4.5.1 Adjacency Destruction Hardness

Main idea. This metric asks how much of the original neighborhood structure survives after the tile permutation. If tiles that were

next to each other are no longer neighbors, the image becomes harder for a convolutional model that relies on local structure.

For every original rightward and downward neighboring relation, we test whether the same two source tiles remain adjacent in the same direction after permutation. The number of such canonical adjacency relations is

$$M = 2N(N - 1).$$

If $A_{\text{preserved}}$ is the number of preserved directed adjacencies, adjacency destruction is

$$A = 1 - \frac{A_{\text{preserved}}}{M}.$$

A low value means that much of the original local structure is preserved. A high value means that neighboring tile relations were mostly destroyed.

4.5.2 Global Tile Displacement

Main idea. This metric measures how far tiles move from their original positions. A permutation that only nudges tiles nearby should preserve more global shape than one that sends tiles across the image.

For each tile, we compute its Manhattan displacement in grid coordinates:

$$\delta_i = |r'_i - r_i| + |c'_i - c_i|.$$

The normalized global displacement is

$$D = \frac{\sum_i \delta_i}{D_{\text{max}}},$$

where D_{max} is the maximum possible total Manhattan displacement for an $N \times N$ grid. A low value indicates that tiles remain close to their original locations. A high value indicates large-scale spatial relocation.

4.5.3 Spatial Permutation Entropy

Main idea. This metric asks whether tile movement follows a simple pattern or many different movement patterns. More varied movement directions and distances mean the permutation is less orderly.

For each tile, we compute

$$\Delta_i = (r'_i - r_i, c'_i - c_i)$$

and assign it to a movement bin

$$b_i = (\text{direction}(\Delta_i), \text{distanceTier}(\Delta_i)).$$

The direction is one of

$\{\text{stationary}, N, S, E, W, NE, NW, SE, SW\}.$

For non-stationary tiles, the normalized Manhattan distance is

$$d_i = \frac{|r'_i - r_i| + |c'_i - c_i|}{2(N - 1)}.$$

The distance tier is near when $0 < d_i \leq 1/3$, medium when $1/3 < d_i \leq 2/3$, and far when $2/3 < d_i \leq 1$. Stationary tiles use a single stationary bin. If p_b is the empirical probability of movement bin b , the normalized entropy is

$$E = \frac{-\sum_b p_b \log p_b}{\log(N^2)}.$$

A low value means that movement is structured or concentrated in a small number of movement patterns. A high value means that the permutation contains diverse movement directions and distances, making it more disordered.

4.5.4 Combined Hardness Score

Main idea. The combined score summarizes the previous metrics into one number so permutations can be ranked by an overall notion of difficulty.

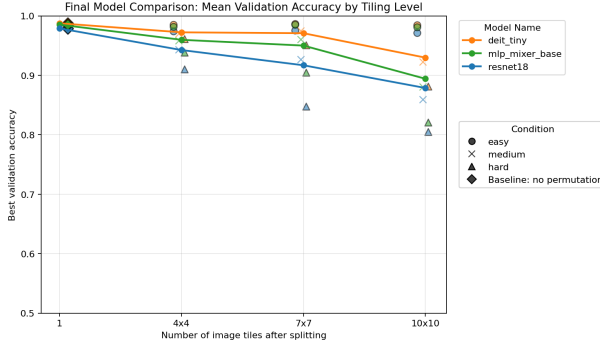


Figure 2: Part 1 validation accuracy as a function of tile count for ResNet-18, DeiT-Tiny, and MLP-Mixer Base.

The final score is a convex combination of the three normalized components:

$$C = w_a A + w_d D + w_e E,$$

with default weights

$$w_a = 0.5, \quad w_e = 0.3, \quad w_d = 0.2.$$

A low combined score represents an easier permutation with preserved structure, small movement, and low movement disorder. A high combined score represents a harder permutation across these complementary axes.

5 Results

Part 1 shows that all three frozen-backbone models remain far above chance under deterministic tile permutations, but performance decreases as the grid becomes more disruptive. DeiT-Tiny is strongest across the matrix, with mean best validation accuracy of 92.97% even at 100 tiles. MLP-Mixer Base is usually second, while ResNet-18 degrades most sharply as the easy, medium, and hard records are averaged at larger grids. The 100-tile mean best accuracy falls to 87.87% for ResNet-18, compared with 89.42% for MLP-Mixer Base and 92.97% for DeiT-Tiny.

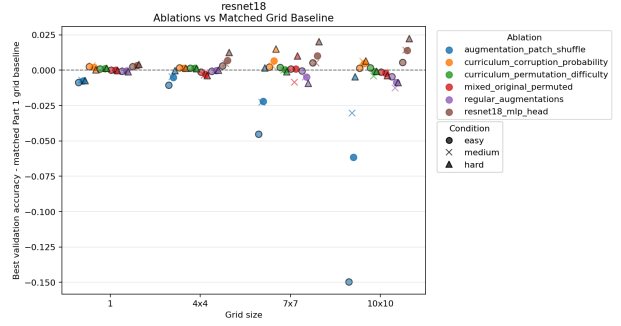


Figure 3: Part 2 best-validation-accuracy delta from the matched ResNet-18 baseline.

Part 2 gives a more selective picture than the earlier exploratory ablations. The ResNet-18 MLP head is the strongest improvement, adding 0.86 percentage points of mean best validation accuracy over the matched baseline. Corruption-probability curriculum is the next-best strategy at +0.39 percentage points. Permutation-difficulty curriculum is approximately neutral on average, while mixed original/permutated training and regular augmentations are slightly below baseline. Patch shuffle is clearly harmful in this final matrix, losing 2.40 percentage points on mean best validation accuracy.

Part 3 provides a ranking lens for the deterministic permutations. All four hardness metrics are negatively correlated with ResNet-18 best validation accuracy over the 12 joined records. Adjacency destruction is the strongest single metric, with Pearson correlation -0.9044 and Spearman correlation -0.9736, indicating that broken local tile-neighborhood structure is especially predictive of lower validation accuracy. The combined hardness score is also strongly negative, but it does not exceed adjacency destruction on this run matrix.

6 Discussion

The proposal asks whether a neural network can classify post-permutation images and whether performance depends on tile resolution and permutation choice. The final deterministic matrix answers the first question affirmatively: every model and every ResNet-18 ablation remains well above chance, including the hard 10-by-10 cases. At the same time, the results show that the task is not trivial. Accuracy drops with stronger deterministic disruption, and the drop is largest for the convolutional ResNet-18 baseline.

The architecture comparison suggests that patch-based pretrained representations are especially effective in this setting. DeiT-Tiny has the best aggregate performance, and MLP-Mixer Base also exceeds ResNet-18 at larger tile counts. This does not prove that attention or token mixing solves permutation hardness in general, because the backbones are frozen and pretrained on natural images, but it does suggest that the relevant class information is not limited to local convolutional continuity.

The Part 2 ablations show that simply adding spatial noise is not enough. Patch shuffle hurts the final matrix, likely because the validation task uses one fixed deterministic permutation rather than arbitrary extra shuffling. The MLP head improves most consistently, suggesting that the frozen ResNet-18 features may contain enough information but benefit from a more expressive classifier. The corruption-probability curriculum gives a smaller positive gain, while the permutation-difficulty curriculum is nearly neutral.

The hardness metrics address the proposal’s third objective: ranking permutations with the same tile count. The strong negative correlations show that geometry-only permu-

tation descriptors capture meaningful variation in model performance. Adjacency destruction is particularly informative, which matches the intuition that destroying local tile neighborhoods is harmful for a convolutional model.

7 Conclusion and Future Work

This project implements a reproducible Dogs versus Cats testbed for studying fixed tile-wise image permutations. The work translates the proposal’s motivation into three concrete outputs: baseline classification under deterministic permutations, improvement ablations for the permuted setting, and permutation hardness metrics joined to validation accuracy. The final implementation standardizes all three parts around four grid settings and three deterministic permutation functions, making the baseline, ablation, and hardness analyses directly comparable.

Future work should expand beyond the current deterministic matrix. The next steps are to repeat experiments across random seeds, compare more architectures in both frozen and fine-tuned regimes, and add transfer tests between permutation functions. These extensions would better separate memorization of a fixed permutation from direct use of class cues in permuted images, which is one of the central questions posed by the original project proposal.

Part 2 strategy	Training set	Added training exposure	Validation set
Regular baseline	Reused Part 1 ResNet-18 runs.	None beyond the fixed target permutation.	Reused Part 1 held-out validation accuracy.
Patch shuffle	Same 20,000 training images and target permutation.	No appended files; each batch of size B produces B shuffled training views, for 20,000 shuffled views per epoch.	Same 5,000 held-out images with the target permutation only.
Regular augmentations	Same 20,000 training images and target permutation.	Standard stochastic training transforms produce one augmented view per example per epoch.	Same 5,000 held-out images with the target permutation only.
Mixed original/permuted	Same 20,000 training images.	Each training view is sampled as original or target-permuted, using $p_{\text{original}} = 0.5$.	Same 5,000 held-out images with the target permutation only.
ResNet-18 MLP head	Same 20,000 training images and target permutation.	No new images; the linear classifier is replaced by a small MLP head.	Same 5,000 held-out images with the target permutation.
Corruption-probability curriculum	Same 20,000 training images at every stage.	No appended files; stage probabilities 0.1, 0.3, 0.5, and 0.8 give expected permuted counts of 2,000, 6,000, 10,000, and 16,000 views per stage epoch.	One fixed target validation loader; no separate validation set per stage.
Permutation-difficulty curriculum	Same 20,000 training images at every stage.	No appended files; each stage processes 20,000 views per epoch at its scheduled difficulty: original, 2-by-2, 3 tiles per side, 4-by-4, and for 7-by-7 or 10-by-10 targets an additional final target-grid stage.	One fixed final-target validation loader; no separate validation set per stage.

Table 2: Part 2 training and validation sets by ablation. The final output matrix uses ResNet-18 with 4 grid settings crossed with easy, medium, and hard deterministic permutation records, for 12 records per ablation.

Part 2 strategy	Stages	Experiment sequence	Dataset increase / per-epoch views
Regular baseline	Part 1 1	Standard training on the target permutation.	No increase: 20,000 training examples per epoch.
Patch shuffle	1	Standard training, with random patch shuffling applied inside training batches.	0 stored increase; B shuffled views per batch and 20,000 per epoch.
Regular augmentations	1	Standard training, with generic image augmentations in the training transform.	0 stored increase; one stochastic view per example per epoch.
Mixed original/permuted	1	Standard training, sampling clean and target-permuted views with equal probability.	0 stored increase; expected 10,000 clean and 10,000 permuted views per epoch.
ResNet-18 head	MLP 1	Standard target-permutation training, replacing the linear head with an MLP head.	No increase: the examples are unchanged, only the trainable head changes.
Corruption-probability curriculum	4	Permutation probability stages: 0.1, 0.3, 0.5, 0.8.	0 stored increase; expected permuted views per epoch are 2,000, 6,000, 10,000, and 16,000 by stage.
Permutation-difficulty curriculum	1, 4, or 5	Baseline target: original only. Tiled targets: original, 2-by-2, 3 tiles per side, 4-by-4, and the final target grid when larger than 4-by-4, using the same easy/medium/hard function as the target record.	0 stored increase; 20,000 scheduled-difficulty views per stage epoch.

Table 3: Part 2 ablation experiment design. Non-curriculum ablations train for 10 epochs; both curriculum ablations train for 30 epochs. None of the ablations duplicate files on disk.



Figure 4: Part 3 example permutations and model-agnostic hardness metrics for the deterministic grid and difficulty settings.

Part 3 analysis	Training set	Validation source
Hardness metric computation	Not applicable; Part 3 does not train a model.	Reuses Part 1 ResNet-18 validation accuracies.
Metric-accuracy join	Not applicable; only saved CSV outputs are joined.	Matches each metric row to the corresponding Part 1 permutation ID.
Correlation analysis	Not applicable; no parameters are updated.	Computes correlations against the joined held-out validation accuracies.

Table 4: Part 3 data usage. Part 3 analyzes saved permutations and validation accuracies rather than creating new train or validation splits.

Model	Main operation	Inductive bias
ResNet-18	Convolution	Local texture and translation structure
DeiT-Tiny	Self-attention	Patch-to-patch interactions
MLP-Mixer Base	MLP mixing	Patch and feature mixing without attention

Table 5: Architectural comparison of the three frozen feature extractors.

Model	Learnable	Frozen	Total
DeiT-Tiny	386	5,524,416	5,524,802
ResNet-18	1,026	11,176,512	11,177,538
MLP-Mixer Base	1,538	59,111,472	59,113,010

Table 6: Parameter counts for the frozen-backbone comparison with two output classes. “Learnable” counts only the classifier head optimized during training; “Frozen” counts pretrained parameters used only in the forward pass.

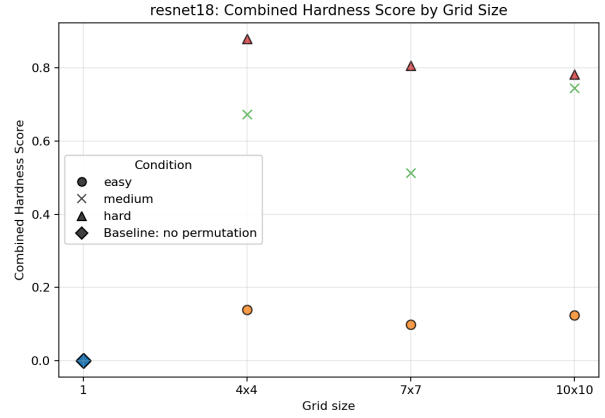


Figure 5: Combined hardness score by grid size and deterministic difficulty label.

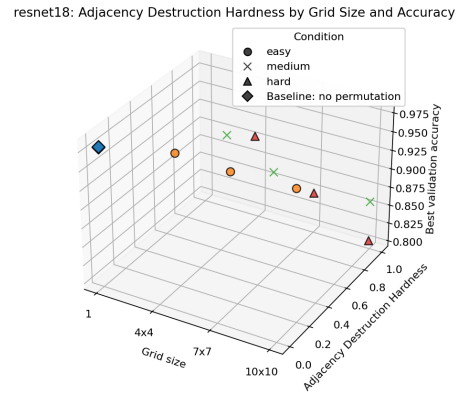


Figure 6: Adjacency destruction, grid size, and ResNet-18 best validation accuracy across the joined deterministic records.

Metric	Dimension	What it captures
Adjacency destruction hardness	Local topology	Are original neighbors still neighbors?
Global tile displacement	Global geometry	How far did tiles move from their original positions?
Spatial permutation entropy	Permutation disorder	How diverse or irregular are tile movement patterns?
Combined hardness score	Aggregated hardness	Weighted summary of the three normalized axes.

Table 7: Tile permutation hardness metrics used in the analysis.

Model	1 tile best	16 tiles best	49 tiles best	100 tiles best
ResNet-18	97.89%	94.25%	91.69%	87.87%
DeiT-Tiny	98.75%	97.23%	97.08%	92.97%
MLP-Mixer Base	98.52%	95.97%	95.00%	89.42%

Table 8: Part 1 mean best validation accuracy by model and tile count, aggregated across the easy, medium, and hard deterministic records at each grid size.

Part 2 setting	Mean final acc.	Mean best acc.	Mean best delta vs. baseline
Regular Part 1 baseline	92.85%	92.92%	0.00 pp
Patch shuffle	89.82%	90.52%	-2.40 pp
Regular augmentations	92.36%	92.58%	-0.35 pp
Mixed original/permuted	92.76%	92.83%	-0.10 pp
ResNet-18 MLP head	93.41%	93.79%	+0.86 pp
Corruption-probability curriculum	93.08%	93.31%	+0.39 pp
Permutation-difficulty curriculum	92.87%	92.98%	+0.05 pp

Table 9: Part 2 ResNet-18 ablation summary over the 12 deterministic records. The delta column compares each record’s best validation accuracy against the matched regular Part 1 ResNet-18 baseline and then averages those deltas.

Hardness metric	Pearson	Spearman
Global displacement	-0.7607	-0.8169
Adjacency destruction	-0.9044	-0.9736
Spatial entropy	-0.6942	-0.8310
Combined hardness	-0.8477	-0.9296

Table 10: Part 3 correlations between ResNet-18 best validation accuracy and permutation metrics over the 12 deterministic records.

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